SIMULOS® Architecture Map

Simulation Governance Architecture — Simulation Governance Layer

SIMULOS® serves as the official OOF® simulation governance reference map used to identify, define, classify, navigate, govern, evaluate, and understand governable simulation spaces across AI systems, autonomous systems, robotics, digital twins, engineering systems, scientific models, critical infrastructure, space systems, and future simulation environments.

The purpose of SIMULOS® is not to define the technical implementation of a simulation system.

The purpose of SIMULOS® is to identify, define, separate, and map the governable spaces through which a Simulation Object is identified, scoped, represented, assumption-bounded, modeled, supplied with inputs, exposed to scenarios, executed, interpreted through outputs, correlated with reality, converted into bounded evidence, and governed throughout its continuing lifecycle.

SIMULOS® serves as the official simulation governance reference map for:

- AI Systems
- Autonomous Systems
- Autonomous Agents
- Robotics
- Digital Twins
- Engineering Systems
- Scientific Models
- Physical Simulations
- Hybrid Simulations
- Mission Simulations
- Space Systems
- Safety-Critical Systems
- Industrial Systems
- Transportation Systems
- Healthcare Simulations
- Financial and Economic Simulations
- Critical Infrastructure
- Multi-System Environments
- Future Simulation Ecosystems

Core Architectural Principle

Every simulation governance space answers one primary governance question.

No simulation governance space duplicates the purpose of another simulation governance space.

Every simulation governance space has a defined beginning, a defined operational purpose, and a defined governance boundary.

Every simulation governance space begins where the preceding simulation governance requirement has been sufficiently established.

Together, these simulation governance spaces form the official simulation governance reference map of SIMULOS®. Official Simulation Governance Flow

Simulation Object → Simulation Purpose & Scope → Simulation Reality Representation → Simulation Assumptions → Simulation Model & Environment → Simulation Inputs & Data → Simulation Scenario Design → Simulation Execution → Simulation Output Governance → Simulation-to-Reality Correlation → Simulation Evidence & Reliance Boundary → Simulation Lifecycle, Drift & Resimulation 1. Simulation Object Standard (SOS)

Governed Space: Simulation Object Core Question

What exactly is being simulated, and how must the Simulation Object be identified, bounded, versioned, and distinguished? Canonical Definition

Simulation Object Standard (SOS) defines the governance conditions under which the object, system, process, environment, phenomenon, configuration, relationship, or other subject being represented through simulation is formally identified, bounded, versioned, configured, distinguished, and documented as a governable Simulation Object. Governance Boundary

Begins when an intended simulation requires explicit identification of what is actually being simulated.

Ends when the Simulation Object has been sufficiently identified, bounded, versioned, configured, and documented to support definition of the simulation purpose and scope. 2. Simulation Purpose & Scope Standard (SPSS)

Governed Space: Simulation Purpose & Scope Core Question

Why is the simulation being performed, what must it cover, and what remains outside its simulation boundary? Canonical Definition

Simulation Purpose & Scope Standard (SPSS) defines the governance conditions under which the purpose, intended inquiry, scope, exclusions, applicability context, required depth, and required precision of a simulation are formally established and documented. Governance Boundary

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Begins when the Simulation Object has been sufficiently established and the reason for simulating it must be formally defined.

Ends when the simulation purpose, scope, exclusions, applicability context, depth, and precision have been sufficiently established to determine which aspects of reality must be represented. 3. Reality Representation Standard (RRS)

Governed Space: Simulation Reality Representation Core Question

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Which aspects of reality must the simulation represent, at what abstraction and fidelity, and which aspects may legitimately remain simplified or excluded? Canonical Definition

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Reality Representation Standard (RRS) defines the governance conditions under which relevant aspects of reality are selected, abstracted, represented, resolved, simplified, limited, or excluded within a simulation according to the simulation's governed purpose and scope. Governance Boundary

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Begins when the simulation purpose and scope are sufficiently established to determine which aspects of reality materially require representation.

Ends when the required reality representation, abstraction, fidelity, resolution, exclusions, and known representational limitations have been sufficiently defined to support explicit governance of simulation assumptions. 4. Simulation Assumption Standard (SAS)

Governed Space: Simulation Assumptions Core Question

Which assumptions enable the simulation, and how must their basis, dependencies, uncertainty, limitations, and consequences remain governed? Canonical Definition

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Simulation Assumption Standard (SAS) defines the governance conditions under which assumptions enabling or materially influencing a simulation are identified, justified, linked to their basis and dependencies, characterized for uncertainty, bounded by limitations, and maintained as explicit elements of the simulation basis. Governance Boundary

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Begins when the selected representation of reality requires assumptions, approximations, inferred conditions, or other propositions that cannot simply

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be treated as established simulation facts.

Ends when material assumptions, their bases, dependencies, uncertainties, and limitations have been sufficiently established to support construction and configuration of the simulation model and environment. 5. Simulation Model & Environment Standard (SMES)

Governed Space: Simulation Model & Environment Core Question

How must the simulation model and simulation environment be constructed, configured, bounded, and controlled? Canonical Definition

Simulation Model & Environment Standard (SMES) defines the governance conditions under which simulation models and simulation environments are structurally constituted, configured, parameterized, bounded, versioned, and controlled to represent the governed Simulation Object under the established simulation basis. Governance Boundary

Begins when the Simulation Object, purpose, reality representation, and material assumptions provide a sufficient basis for model and environment construction.

Ends when the simulation model, environment, parameters, boundary interactions, configuration, and version have been sufficiently established to govern the inputs and data entering the simulation. 6. Simulation Input & Data Standard (SIDS)

Governed Space: Simulation Inputs & Data Core Question

Which inputs and data may enter the simulation, and under what representativeness, transformation, temporal, and uncertainty conditions? Canonical Definition

Simulation Input & Data Standard (SIDS) defines the governance conditions under which simulation inputs and data are identified, selected, assessed for representativeness, transformed, temporally situated, uncertainty-characterized, and admitted into a governed simulation. Governance Boundary

Begins when the configured simulation model and environment require governed inputs, parameters, datasets, signals, initial conditions, or other simulation-driving information.

Ends when the simulation input basis has been sufficiently identified, represented, transformed, temporally bounded, and uncertainty-characterized to support governed scenario design. 7. Simulation Scenario Standard (SSS)

Governed Space: Simulation Scenario Design Core Question

Which scenarios must be simulated to represent expected conditions, variations, edge cases, stresses, uncertainty, and materially relevant alternatives? Canonical Definition

Simulation Scenario Standard (SSS) defines the governance conditions under which simulation scenarios are identified, varied, structured, stress-tested, extended toward relevant edge cases, and evaluated for coverage against the governed simulation purpose and represented reality. Governance Boundary

Begins when the Simulation Object, represented reality, assumptions, model, environment, and input basis are sufficiently established to determine which conditions and variations require simulation.

Ends when the required scenario space, variations, edge cases, stress conditions, and coverage have been sufficiently established to support governed simulation execution. 8. Simulation Execution Standard (SES)

Governed Space: Simulation Execution Core Question

How must a simulation be instantiated, executed, observed, repeated, controlled, and documented? Canonical Definition

Simulation Execution Standard (SES) defines the governance conditions under which a simulation is instantiated, controlled, executed, observed, repeated, and documented so that the execution process remains connected to its approved Simulation Object, configuration, inputs, scenarios, and simulation basis. Governance Boundary

Begins when the simulation object, model, environment, inputs, and required scenario conditions are sufficiently established for execution.

Ends when governed simulation executions have been completed, observed, repeated where required, and sufficiently documented to establish identifiable simulation outputs. 9. Simulation Output Governance Standard (SOGS)

Governed Space: Simulation Output Governance Core Question

What exactly did the simulation produce, how must simulation outputs be bounded and interpreted, and what may legitimately be inferred from them? Canonical Definition

Simulation Output Governance Standard (SOGS) defines the governance conditions under which simulation outputs are identified, classified, interpreted, bounded, documented, and distinguished from derived inferences so that the meaning and limitations of what the simulation produced remain explicit. Governance Boundary

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Begins when governed simulation execution produces observable, measurable, derived, aggregated, or otherwise identifiable outputs.

Ends when simulation outputs and governed inferences have been sufficiently identified, classified, bounded, limited, and documented to support comparison with the reality they claim to represent. 10. Simulation-to-Reality Correlation Standard (SRCS)

Governed Space: Simulation-to-Reality Correlation Core Question

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To what extent do simulated behavior, conditions, and outputs correspond to the reality they claim to represent? Canonical Definition

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Simulation-to-Reality Correlation Standard (SRCS) defines the governance conditions under which simulated behavior, states, conditions, relationships, and outputs are systematically compared with available reality to determine correspondence, deviation, alignment, uncertainty, and the limits of simulation-to-reality correlation. Governance Boundary

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Begins when governed simulation outputs exist and a relevant reality basis is available or can legitimately be established for comparison.

Ends when the simulation-to-reality relationship has been sufficiently characterized through comparison, correlation criteria, deviation analysis, alignment assessment, and correlation uncertainty to support evaluation of simulation-derived evidence.

Simulation similarity is not reality equivalence. 11. Simulation Evidence & Reliance Boundary Standard (SERBS)

Governed Space: Simulation Evidence & Reliance Boundary Core Question

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What evidential meaning may simulation results legitimately carry, and where does simulation-specific reliance end before broader validation governance becomes necessary? Canonical Definition

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Simulation Evidence & Reliance Boundary Standard (SERBS) defines the governance conditions under which simulation-derived information may legitimately qualify as evidence, acquire simulation-specific evidential weight, support explicitly bounded reliance, remain subject to defined use

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limitations, and enter broader validation governance without losing the conditions that establish its legitimate meaning. Governance Boundary

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Begins when governed simulation outputs and, where relevant, simulation-to-reality correlation provide a basis from which simulation-derived evidence may be considered.

Ends when the simulation-derived evidence has been sufficiently qualified, weighted, bounded for reliance, constrained by applicable use limitations, and prepared for transfer into broader validation governance where required.

Simulation-derived evidence must not travel farther than its governed meaning can legitimately support. 12. Simulation Lifecycle & Resimulation Standard (SLRS)

Governed Space: Simulation Lifecycle, Drift & Resimulation Core Question

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How must simulation relevance be monitored over time, and when must changed reality trigger update, resimulation, replacement, or retirement? Canonical Definition

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Simulation Lifecycle & Resimulation Standard (SLRS) defines the governance conditions under which the continuing relevance of a simulation is monitored, material simulation drift is recognized, resimulation triggers are established, resimulation scope is determined, and simulations are updated, repeated, replaced, transitioned, or retired as the Simulation Object, represented reality, assumptions, models, inputs, scenarios, operational conditions, evidence basis, or intended reliance materially change. Governance Boundary

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Begins when a governed simulation and its outputs, correlations, or evidence enter continuing use, reference, reliance, monitoring, or preservation beyond the initial simulation process.

Continues throughout the remaining simulation lifecycle while the simulation, its outputs, or simulation-derived evidence retain operational, evidential, scientific, engineering, analytical, or decision relevance.

Where material change invalidates or weakens the existing simulation basis, Simulation Lifecycle & Resimulation Standard may trigger governed re-entry into the appropriate earlier SIMULOS® governance space rather than requiring the entire lifecycle to restart automatically. Core Governed Simulation Spaces

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- Simulation Object
- Simulation Purpose & Scope
- Simulation Reality Representation
- Simulation Assumptions

- Simulation Model & Environment
- Simulation Inputs & Data
- Simulation Scenario Design
- Simulation Execution
- Simulation Output Governance
- Simulation-to-Reality Correlation
- Simulation Evidence & Reliance Boundary
- Simulation Lifecycle, Drift & Resimulation

Recursive Simulation Lifecycle

SIMULOS® does not end when a simulation execution is completed.

Once simulation results, simulation-derived evidence, or simulation-supported decisions remain in use, the architecture becomes recursive:

Governed Simulation → Simulation Output → Simulation-Derived Evidence → Permitted Reliance → Lifecycle Monitoring → Material Change → Simulation Drift Recognition → Resimulation Trigger → Resimulation Scope → Lifecycle Re-Entry → Updated Simulation / Resimulation / Replacement / Retirement

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Lifecycle re-entry occurs at the earliest materially affected governance space.

A changed Simulation Object may require re-entry into Simulation Object Standard.

A changed purpose may require re-entry into Simulation Purpose & Scope Standard.

A changed understanding of reality may require re-entry into Reality Representation Standard.

A failed assumption may require re-entry into Simulation Assumption Standard.

A changed model or environment may require re-entry into Simulation Model & Environment Standard.

Changed data may require re-entry into Simulation Input & Data Standard.

A newly relevant operating condition may require re-entry into Simulation Scenario Standard.

The architecture therefore supports targeted governed resimulation rather than uncontrolled repetition. Architectural Position

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SIMULOS® governs the complete simulation governance lifecycle.

The architecture progresses through:

Simulation Object → Simulation Purpose & Scope → Simulation Reality Representation → Simulation Assumptions → Simulation Model & Environment → Simulation Inputs & Data → Simulation Scenario Design → Simulation Execution → Simulation Output Governance → Simulation-to-Reality Correlation → Simulation Evidence & Reliance Boundary → Simulation Lifecycle, Drift & Resimulation

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Together, these simulation governance spaces govern how the object of simulation is formally established, why simulation is performed, how reality is represented, which assumptions enable that representation, how models and environments are constituted, which inputs enter the simulation, which scenarios are examined, how simulations are executed, what their outputs legitimately mean, how those outputs correspond to reality, what evidential significance and reliance they may support, and when changing conditions require governed resimulation, replacement, transition, or retirement.

SIMULOS® therefore provides a complete governance path from the first identification of what is being simulated to the continuing determination of whether that simulation remains relevant enough to be relied upon. What SIMULOS® Defines

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SIMULOS® defines:

- where simulation governance begins, - how a Simulation Object is formally identified and bounded, - why a simulation is being performed, - what the simulation must and must not cover, - which aspects of reality must be represented, - what abstraction, fidelity, and resolution are appropriate, - which assumptions enable or materially affect the simulation, - how the simulation model and environment are governed, - which inputs and data may enter the simulation, - which scenarios, variations, edge cases, and stress conditions require simulation, - how simulation execution is governed, - how simulation outputs are identified and interpreted, - where output interpretation ends and inference begins, - how simulated behavior is compared with reality, - how simulation-to-reality deviation and uncertainty are represented, - what simulation-derived information may legitimately become evidence, - how much simulation-specific evidential weight that evidence may carry, - where legitimate simulation-specific reliance ends, - which uses and interpretations must remain limited, - how simulation-derived evidence interfaces with VALIDOS® validation governance, - how simulation relevance is monitored over time, - what constitutes material simulation drift, - when resimulation must be triggered, - where lifecycle re-entry must occur, - when a simulation should be updated, replaced, transitioned, or retired.

SIMULOS® provides the official simulation governance language used for simulation-space identification, simulation lifecycle navigation, governance classification, structural diagnosis, evidence-boundary determination, resimulation governance, and simulation architecture design across complex operational environments. Critical Architectural Distinctions

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Simulation Object ≠ Validation Object

Represented Reality ≠ Operational Reality

Simulation Assumption ≠ Established Fact

Representation Fidelity ≠ Reality Equivalence

Model Sophistication ≠ Model Applicability

Input Volume ≠ Input Representativeness

Scenario Count ≠ Scenario Coverage

Simulation Execution ≠ Audit of Simulation Execution

Simulation Output ≠ Simulation Evidence

Simulation Inference ≠ Direct Simulation Observation

Simulation-to-Reality Correlation ≠ Reality Equivalence

Evidence Volume ≠ Evidential Strength

Repeated Simulation ≠ Independent Evidence

High Evidential Weight ≠ Broad Reliance

Simulation Evidence ≠ Validation Determination

Past Simulation Relevance ≠ Continuing Simulation Relevance

Resimulation ≠ Failure of the Original Simulation Cross-Architecture Governance Boundaries SIMULOS® ↔ ORA™ — Operational Reality Architecture

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Operational Reality Architecture governs what exists and occurs in actual Operational Reality.

SIMULOS® — Simulation Governance Architecture governs how selected aspects of that reality are represented, modeled, simulated, compared, and governed inside the simulation lifecycle.

Operational Reality ≠ Simulated Reality. SIMULOS® ↔ VALIDOS® — Validation Governance Architecture

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SIMULOS® — Simulation Governance Architecture determines what simulation-derived evidence can legitimately represent, mean, and support within its simulation-specific reliance boundary.

VALIDOS® — Validation Governance Architecture determines whether the complete validation basis is sufficient to support a Validation Determination and intended reliance.

Simulation Evidence ≠ Validation Determination. SIMULOS® ↔ INTEGROS® — Integrity Governance Architecture

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SIMULOS® — Simulation Governance Architecture governs simulation-specific structure, conditions, execution, outputs, correlation, evidence boundaries, and lifecycle.

INTEGROS® — Integrity Governance Architecture governs integrity across governed objects, structures, conditions, states, evidence, and processes.

Simulation Governance ≠ Integrity Governance. SIMULOS® ↔ ADIT® — Audit in Real Time Architecture

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SIMULOS® — Simulation Governance Architecture governs simulation execution and the simulation lifecycle itself.

ADIT® — Audit in Real Time Architecture governs evidential observation, preservation, verification, traceability, and auditability of what occurred where applicable.

Execution Governance ≠ Audit Governance. SIMULOS® ↔ AGA™ — Accountability Governance Architecture

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SIMULOS® — Simulation Governance Architecture governs the methodological conditions under which simulation activities and simulation-derived evidence acquire legitimate meaning.

AGA™ — Accountability Governance Architecture governs who holds authority, responsibility, accountability, attribution, and decision ownership surrounding those activities.

Methodological Governance ≠ Accountability Governance. SIMULOS® ↔ AIG® — AI Governance Architecture

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SIMULOS® may simulate AI systems and their behavior.

AIG® — AI Governance Architecture continues to govern the behavioral governance lifecycle of actual AI systems.

Simulated AI Behavior ≠ Governance of Actual AI Behavior. SIMULOS® ↔ ASGA™ — Autonomous Systems Governance Architecture

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SIMULOS® may simulate autonomous agents, robots, vehicles, machines, and multi-agent systems.

ASGA™ — Autonomous Systems Governance Architecture continues to govern the authority, autonomy, constraints, execution, coordination, escalation, and operational behavior of actual autonomous systems.

Simulated Autonomy ≠ Governed Operational Autonomy. Compatible OOF® Architectures

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- GOA™ — Governance Architecture
- ORA™ — Operational Reality Architecture
- VALIDOS® — Validation Governance Architecture
- INTEGROS® — Integrity Governance Architecture
- OBIDENITY® — Origin-Bound Identity Governance Architecture
- AGA™ — Accountability Governance Architecture
- AIG® — AI Governance Architecture
- ASGA™ — Autonomous Systems Governance Architecture
- CLIA® — Cognitive Governance Intelligence Architecture
- MGIA™ — Memory Governance Intelligence Architecture
- ADIT® — Audit in Real Time Architecture

Canonical Principle

SIMULOS® does not define Operational Reality, general validation sufficiency, general integrity, accountability, auditability, or the operational authority and behavior of actual AI and autonomous systems. SIMULOS® governs the conditions under which a Simulation Object is identified, simulation purpose is bounded, reality is represented, assumptions are made explicit, models and environments are constituted, inputs are admitted, scenarios are designed, simulations are executed, outputs are interpreted, simulation-to-reality relationships are established, simulation-derived evidence acquires bounded meaning and reliance, and material change triggers governed lifecycle re-entry, resimulation, replacement, transition, or retirement. Canonical Closing Statement

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SIMULOS® serves as the official simulation governance reference map for AI systems, autonomous systems, robotics, digital twins, engineering systems, scientific models, critical infrastructure, space systems, and future simulation environments. By defining the primary governable spaces of the complete simulation lifecycle, SIMULOS® provides the architecture used to identify what is being simulated, establish why it is being simulated, determine how reality is represented, govern assumptions, models, environments, inputs, scenarios and execution, establish what simulation outputs legitimately mean, determine how simulated results correspond to reality, bound the evidential meaning and reliance of simulation-derived evidence, and continuously determine whether material change requires lifecycle re-entry, resimulation, replacement, transition, or retirement. SIMULOS® therefore establishes simulation not as an isolated computational event, but as a continuously governable relationship between a Simulation Object, a representation of reality, a governed simulation process, simulation-derived evidence, and the changing reality within which that evidence may ultimately be relied upon.

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Canonical Source

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